

BROOKS COUNTY AP, TX, USA

WMO: 720269

Lat: 27.207N

Lon: 98.121W

Elev: 34

StdP: 100.92

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12982

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.4	2.9	-8.8	1.8	12.7	-6.4	2.2	13.0	11.0	24.0	9.4	22.1	2.3	350	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	38.7	24.0	37.6	24.1	36.7	24.1	27.1	32.1	26.6	31.7	26.2	31.3	5.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.6	28.6	25.4	20.7	28.2	25.1	20.3	28.0	85.0	32.1	83.0	31.6	81.4	31.0	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.3	7.5	DB	-0.7	40.8	1.6	1.8	-1.8	42.1	-2.7	43.1	-3.6	44.1	-4.7	45.4	
			WB	-2.8	27.7	1.6	0.7	-3.9	28.3	-4.8	28.7	-5.7	29.1	-6.9	29.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	14.8	17.6	20.7	24.1	27.2	29.4	29.6	30.0	28.0	24.4	19.4	15.9
	DBStd	6.64	5.16	5.58	4.71	3.73	2.94	1.78	2.29	2.34	2.16	3.79	5.01	5.29
	HDD10.0	38	16	7	2	0	0	0	0	0	0	3	10	
	HDD18.3	408	133	74	32	4	0	0	0	0	5	48	113	
	CDD10.0	4950	165	221	334	423	534	582	606	619	540	447	285	192
	CDD18.3	2279	23	54	106	177	276	332	348	361	290	194	80	37
	CDH23.3	24808	207	425	841	1680	2959	3982	4300	4525	3022	1907	675	284
	CDH26.7	12214	40	134	299	730	1428	2119	2339	2541	1474	836	209	63
Wind	WSAvg	3.2	3.0	3.4	3.7	3.9	3.9	3.6	3.5	3.1	2.4	2.7	2.8	3.0
Precipitation	PrecAvg	658	30	38	23	34	75	80	45	69	113	80	34	31
	PrecMax	1401	97	157	119	138	315	330	294	327	833	263	133	127
	PrecMin	319	0	0	0	0	0	0	0	12	0	0	0	0
	PrecStd	232	24	35	24	35	62	77	56	75	126	68	29	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	32.8	35.0	37.2	38.0	39.2	40.0	40.5	38.0	35.7	32.7	30.7
		MCWB	19.0	20.4	20.0	22.9	23.7	23.9	23.4	24.7	22.8	24.0	22.6	20.8
	2%	DB	27.6	30.2	31.9	34.9	36.6	37.5	38.2	38.8	36.4	33.9	31.0	28.3
		MCWB	18.5	19.7	21.1	22.0	23.6	24.1	24.0	24.5	23.6	23.8	22.0	19.8
	5%	DB	25.9	28.2	30.0	32.6	35.1	36.3	37.1	37.6	35.0	32.6	29.1	26.6
		MCWB	17.6	19.4	20.2	21.5	23.8	24.2	24.3	24.5	24.4	23.3	21.1	19.0
	10%	DB	23.6	26.1	27.9	31.1	33.2	35.1	35.6	36.3	33.4	31.3	27.4	24.6
		MCWB	17.1	18.3	19.7	21.4	23.3	24.4	24.5	24.7	24.3	23.0	20.2	19.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	22.4	23.3	25.8	26.8	27.4	27.5	28.0	27.3	26.7	24.5	22.8
		MCDB	25.3	28.2	29.3	32.0	33.5	33.6	31.7	33.6	31.7	31.0	29.1	26.3
	2%	WB	20.8	21.7	22.5	24.7	25.8	26.6	26.8	27.1	26.6	25.8	23.7	21.9
		MCDB	24.2	26.5	27.5	30.3	31.7	32.3	31.5	32.1	31.1	30.3	27.9	25.4
	5%	WB	19.9	21.1	21.9	23.9	25.4	26.1	26.2	26.5	26.1	25.2	22.8	21.2
		MCDB	23.9	25.3	26.3	29.2	30.7	31.5	31.2	31.4	30.6	29.5	26.7	24.8
	10%	WB	18.8	20.4	21.4	23.2	24.9	25.7	25.8	26.1	25.6	24.6	22.0	20.3
		MCDB	22.6	24.4	25.5	27.9	30.0	30.8	30.7	31.1	30.2	28.9	25.9	23.6
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.2	12.2	12.7	11.8	11.8	12.1	12.9	11.1	12.5	12.4	11.9
		MCDBR	15.2	14.1	13.7	14.1	13.5	13.4	13.8	14.3	13.2	12.7	13.1	14.2
		MCWBR	7.7	5.8	5.1	5.0	3.5	3.1	3.1	3.2	3.6	4.4	5.5	7.1
	5% WB	MCDBR	11.5	11.4	11.0	11.3	10.8	11.2	11.4	12.3	10.5	10.8	10.9	11.3
	MCWBR	6.8	5.2	4.8	4.3	3.4	3.2	3.3	3.4	3.5	4.6	5.5	6.9	
Clear-Sky Solar Irradiance	taub	0.354	0.368	0.385	0.422	0.455	0.438	0.437	0.437	0.427	0.379	0.367	0.348	
	taud	2.453	2.408	2.369	2.280	2.224	2.329	2.342	2.349	2.344	2.470	2.448	2.482	
	Ebn at Noon	891	905	907	878	845	854	853	854	854	883	868	881	
	Edh at Noon	98	110	121	136	144	129	127	125	122	102	97	91	
All-Sky Solar Radiation	RadAvg	3.10	3.73	4.62	5.49	5.97	6.67	6.63	6.37	5.24	4.55	3.46	2.79	
	RadStd	0.35	0.62	0.56	0.54	0.50	0.29	0.51	0.33	0.45	0.50	0.38	0.41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	+0.34	N/A	N/A	N/A	N/A	+129	+117

Nomenclature: See separate page